

Literature status: superreflexive Δ -point subcases of Question 7.7

Run `fa.banach.001`; source arXiv:2301.04433

Packet Status

This packet records a literature-already-answered partial subcase, not an original construction from the run. The supporting paper explicitly cites Martin–Perreau–Rueda Zoca Question 7.7 and constructs a superreflexive example with strong Δ -type points. The packet does not answer the Daugavet or super Daugavet parts of the source question.

Source Question

The source is Miguel Martin, Yoel Perreau, and Abraham Rueda Zoca, *Diametral notions for elements of the unit ball of a Banach space*, arXiv:2301.04433. Question 7.7 asks for isomorphic restrictions on Banach spaces containing Δ -points or the other diametral notions, and then asks in particular whether a reflexive, or even super-reflexive, Banach space can contain Δ -, super Δ -, ccs Δ -, Daugavet, or super Daugavet points.

By now, the only *isomorphic* restriction which is known for a Banach space to contain Δ -points or even Daugavet points is that it cannot be finite-dimensional. It would be interesting to find some more.

Question 7.7. Find isomorphic restrictions for a Banach space to contain Δ -points or any of the other diametral notions. In particular, is it possible for a reflexive or even super-reflexive Banach space to contain Δ -, super Δ -, ccs Δ -, Daugavet or super Daugavet points?

The results about absolute sums in Subsection 3.2 are not complete in the case of super Daugavet points and they are even less clear in the case of ccs notions. Here are two possible questions.

Supporting Literature

The supporting paper is Trond A. Abrahamsen, Ramon J. Aliaga, Vegard Lima, Andre Martiny, Yoel Perreau, Antonin Prochazka, and Triinu Veeorg, *Delta-points and their implications for the geometry of Banach spaces*, arXiv:2303.00511. Its introduction states that Question 7.7 of Martin–Perreau–Rueda Zoca asks whether finite-dimensional nonexistence of Δ -points also holds for superreflexive spaces, and says this is answered negatively in Section 3.

The key supporting theorem is Theorem 3.1.

more, as we will show that e_1 belongs to the closure of the set of all strongly exposed points of B_Y . This will be done in the lemmas that follow the main theorem of this section that we will now state.

Theorem 3.1. *Let Y be the renorming of ℓ_2 above. We have that both $e_1 \in S_Y$ and $e_1^* \in S_{Y^*}$ are super Δ -points, but neither of them is a Daugavet point.*

The proof of the theorem uses a few simple lemmas. In [DKR⁺16], explicit expressions

The same section then records that the points are actually ccw Δ -points.

Using notation from [MPRZ23] we can say more about e_1 and e_1^* . If C is a convex set, then a given subset D of C is a *convex combination of relatively weakly open subsets* (ccw for short) of C , if D is of the form

$$D := \sum_{i=1}^n \lambda_i W_i,$$

where $n \in \mathbb{N}$, $\lambda_i \in (0, 1]$ for $i = 1, \dots, n$, $\sum_{i=1}^n \lambda_i = 1$ and W_i are relatively weakly open subsets of C .

Then we have that e_1 and e_1^* are not only super Δ -points, but actually ccw Δ -points in the sense of [MPRZ23].

Corollary 3.6. *If C is a ccw of B_Y such that $e_1 \in C$, then $\sup_{y \in C} \|e_1 - y\| = 2$. Similarly, if D is a ccw of B_{Y^*} such that $e_1^* \in D$, then $\sup_{y^* \in D} \|e_1^* - y^*\| = 2$.*

Idea of the Proof

The source question asks whether even superreflexive spaces can support these diametral point notions. The supporting paper builds an equivalent norm on ℓ_2 . Equivalent renorming preserves the isomorphic property of superreflexivity. The supporting theorem then marks e_1 as a super Δ -point, and the following corollary upgrades it to ccw Δ . Since every slice is a relatively weakly open subset, ccw Δ implies ccs Δ ; and super Δ implies ordinary Δ .

Formal Statement

Theorem. *There is a superreflexive Banach space containing a point that is simultaneously a Δ -point, a super Δ -point, and a ccs Δ -point. Consequently, Question 7.7 of arXiv:2301.04433 has an affirmative answer for these three diametral notions.*

Proof. Let Y be the equivalent renorming of ℓ_2 constructed in Section 3 of arXiv:2303.00511. Since Y is linearly isomorphic to Hilbert space ℓ_2 , it is superreflexive.

Theorem 3.1 of arXiv:2303.00511 states that $e_1 \in S_Y$ is a super Δ -point. In particular, e_1 is a Δ -point, because slices containing e_1 are special cases of relatively weakly open subsets of B_Y containing e_1 .

Corollary 3.6 of the same paper states that e_1 is a ccw Δ -point: for every convex combination of relatively weakly open subsets C of B_Y with $e_1 \in C$, one has

$$\sup_{y \in C} \|e_1 - y\| = 2.$$

Every convex combination of slices is a convex combination of relatively weakly open subsets. Hence e_1 is also a ccs Δ -point.

Thus the superreflexive space Y gives affirmative examples for the Δ , super Δ , and ccs Δ parts of Question 7.7. $\square$

Limitations

This is only a partial subcase of Question 7.7. It does not answer whether a reflexive or superreflexive Banach space can contain a Daugavet point or a super Daugavet point. Indeed, Theorem 3.1 of the supporting paper explicitly states that the points $e_1 \in S_Y$ and $e_1^* \in S_{Y^*}$ are not Daugavet points.

Literature And Duplicate Check

The run lightweight indexes were searched for arXiv:2301.04433, diametral notions, Δ -points, Daugavet points, ccs Δ -points, and superreflexive/reflexive keywords. No existing packet, attempt, or proof-gap entry covered Question 7.7. The only nearby registry hit for arXiv:2303.00511 was a different full packet concerning weak-star Δ -points in duals of subspaces with shrinking k -unconditional bases.

Human Review Recommendation

Review as a literature-already-answered partial subcase. The main checks are: the identification of Y as a superreflexive renorming of ℓ_2 , the implication ccw $\Delta \Rightarrow$ ccs Δ , and the stated scope exclusion for Daugavet and super Daugavet points.

References

- [1] M. Martin, Y. Perreau, and A. Rueda Zoca, *Diametral notions for elements of the unit ball of a Banach space*, arXiv:2301.04433.
- [2] T. A. Abrahamsen, R. J. Aliaga, V. Lima, A. Martiny, Y. Perreau, A. Prochazka, and T. Veeorg, *Delta-points and their implications for the geometry of Banach spaces*, arXiv:2303.00511.